

Quantum Nanostructures and Quantum Confinement

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Abstract

Quantum nanostructures are systems where at least one dimension lies in the nanometer range, leading to quantum mechanical effects dominating their behavior. Quantum confinement results in discrete energy levels and modified optical and electronic properties. This document provides detailed derivations, formulas, and physical interpretations.

1 Introduction

When the size of a material becomes comparable to the de Broglie wavelength of electrons, classical physics fails. Instead, quantum mechanics governs behavior, resulting in quantized energy levels and size-dependent properties.

2 Quantum Confinement: Mathematical Foundation

2.1 Time-Independent Schrödinger Equation

The fundamental equation governing quantum systems is:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(x)\psi = E\psi \tag{1}$$

Meaning:

- ψ = wavefunction (probability amplitude of finding a particle)
- $|\psi|^2$ = probability density
- $\hbar$ = reduced Planck's constant ($h/2\pi$)
- m = particle mass
- $V(x)$ = potential energy
- E = total energy

This equation determines allowed energy states and wavefunctions.

3 Particle in a 1D Infinite Potential Well

3.1 Model Setup

A particle confined in a box of length L :

$$[V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{elsewhere} \end{cases}]$$

3.2 Solution

Solving Schrödinger equation gives:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (2)$$

Meaning:

- Only specific standing wave patterns are allowed
- n is the quantum number ($n = 1, 2, 3, \dots$)

3.3 Energy Levels

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (3)$$

Interpretation:

- Energy is quantized (discrete)
- $E \propto \frac{1}{L^2}$: Smaller size $\Rightarrow$ higher energy
- Explains size-dependent properties in nanostructures

4 3D Quantum Confinement (Quantum Dot)

For a cubic box:

$$E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right) \quad (4)$$

Meaning:

- Energy depends on three quantum numbers
- Leads to discrete atomic-like spectra

5 Effective Mass Approximation

In semiconductors:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2m^* L^2} \quad (5)$$

Explanation:

- m^* = effective mass (accounts for crystal lattice effects)
- Electrons behave as if they have a different mass inside materials

6 Exciton and Confinement

6.1 Exciton Bohr Radius

$$a_B = \frac{4\pi\epsilon\hbar^2}{\mu e^2} \quad (6)$$

Meaning:

- μ = reduced mass of electron-hole pair
- ϵ = dielectric constant
- Determines confinement regime

7 Bandgap Shift Due to Quantum Confinement

For semiconductor nanocrystals:

$$E(R) = E_g + \frac{\hbar^2 \pi^2}{2R^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) - \frac{1.8e^2}{4\pi\epsilon R} \quad (7)$$

Explanation:

- $E(R)$ = size-dependent bandgap
- E_g = bulk bandgap
- First term = confinement energy (increases bandgap)
- Second term = Coulomb attraction (reduces energy)

8 Density of States (DOS)

8.1 3D (Bulk)

$$D(E) \propto \sqrt{E} \quad (8)$$

8.2 2D (Quantum Well)

$$D(E) = \text{constant (step-like)} \quad (9)$$

8.3 1D (Quantum Wire)

$$D(E) \propto \frac{1}{\sqrt{E}} \quad (10)$$

8.4 0D (Quantum Dot)

$$D(E) = \sum \delta(E - E_n) \quad (11)$$

Meaning:

- δ = discrete energy spikes
- Reflects atomic-like states

9 Optical Transition Energy

$$E_{\text{photon}} = E_c - E_v \quad (12)$$

Meaning:

- Determines emitted photon energy
- Smaller size $\Rightarrow$ larger E_{photon} (blue shift)

10 Heisenberg Uncertainty Principle

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad (13)$$

Interpretation:

- Confinement (small Δx) $\Rightarrow$ large momentum uncertainty
- Leads to increased kinetic energy

11 Applications

- LEDs and lasers
- Quantum computing (qubits)
- Solar cells
- Biomedical imaging

Questions and Answers

Q1. Why does energy increase when size decreases? Due to confinement, the wavefunction is restricted, increasing kinetic energy ($E \propto 1/L^2$).

Q2. What does the Schrödinger equation represent? It describes how quantum states evolve and determines allowed energies.

Q3. What is the physical meaning of a wavefunction? Its square gives probability density.

Q4. What is an exciton? A bound state of an electron and a hole.

Q5. Why do quantum dots behave like atoms? Because of discrete energy levels from 3D confinement.

Q6. What causes blue shift in nanostructures? Increase in bandgap due to confinement.

Q7. What is effective mass? It represents how electrons respond to forces in a crystal.

Q8. What is density of states? Number of available energy states per energy interval.